

SUBHANSHU SETHI

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Professional Summary: Computer Vision & AI Engineer with 1+ year experience deploying real-time RGB-D and LiDAR perception pipelines on edge hardware (Jetson Nano, Intel NUC Ultra). Specialized in YOLOv8, OpenVINO, TensorRT & ROS. IJCNN 2025 publication + multiple top-tier wins in AI vision systems.

WORK EXPERIENCE

Computer Vision Engineer (AI/ML)

Novus Hi-Tech, Gurugram

Jul 2025 – Present

- **Deployed** real-time RGB-D pose estimation pipeline in **C++/ROS**, achieving **25–30 Hz** inference via **OpenVINO** on Intel NUC Ultra (**CPU, iGPU, NPU**).
- **Engineered** Reflector-Based Localization (RBL) system with **LiDAR** clustering and triangulation, achieving **5–10 cm** pose accuracy in repetitive environments.
- **Optimized** pallet detection by migrating from **EfficientDet** to **YOLOv8** via **TensorRT** on Jetson Nano, delivering **3x** throughput (5–7 to 15–20 FPS).
- **Automated** ML training and versioning workflows using **DVC** to accelerate annotation and model staging.

Computer Vision Intern

Novus Hi-Tech, Gurugram

Jun 2024 – Jun 2025

- **Benchmarked** detection (**YOLO**, **EfficientDet**, **DETR**) and segmentation (**Mask R-CNN**, **SAM**) models for real-time **edge-robotics** performance.
- **Optimized** pose estimation using **RGB-D** depth data (eliminated **RANSAC**) for real-time edge execution across full robotics stack.

EDUCATION

B.Tech in Electrical Engineering

Delhi Technological University

Courses: Deep Learning, Computer Vision, Computer Networks, Cloud Computing, IoT

2021 – 2025

PUBLICATIONS

- **GExSent: Gated Experts for Robust Sentiment Analysis Across Modalities** — IJCNN 2025, Rome, Italy
- **ContXCLIP: Contextual Attention for Vision-Language Understanding** — Under Review, *Signal Processing: Image Communication* (Elsevier)

PROJECTS

GExSent — Multimodal Classification | [GitHub](#)

Feb 2025 – Apr 2025

Tech Stack: PyTorch, CLIP, ModernBERT, Mixture of Experts

- **Achieved** SOTA on MOSI, MOSEI, Memotion (**MAE 0.865**, **F1 87.24**) with **9M**-parameter MoE pipeline (IJCNN 2025 accepted).
- Designed hierarchical gated MoE and attention for multi-modal fusion.

ContXCLIP — Image Captioning | [GitHub](#)

Jul 2024 – Feb 2025

Tech Stack: PyTorch, Vision-Language Models (CLIP), GPT-2

- **Outperformed** MS-COCO baselines (**CIDEr 134.1**, **BLEU-4 39.4**) with **11.7M**-parameter CLIP+GPT-2 architecture.
- Built context preservation (CPM) and cross-modal attention (XCA) modules.

Financial News Sentiment System | [GitHub](#)

Nov 2025 – Dec 2025

Tech Stack: AWS (Lambda, ECS/ECR), FinBERT, OpenAI LLM, Streamlit

- **Reduced** API costs by **65x** and call volume by **96%** via batch processing of 40+ RSS feeds.
- **Deployed** serverless pipeline analyzing 100+ articles in 2–4 seconds.

ROS Agentic Debugger

Mar 2026 – Apr 2026

Tech Stack: ROS, Python, LLMs (Gemma4 4b, Ollama)

- **Developed** ROS debugger using Gemma LLM to automate validation of topics, TF trees, and node flows.
- **Enabled** instant diagnostics on autonomous platforms, eliminating manual checks.

ACHIEVEMENTS

- **Top 15 out of 19,000+** (Flipkart Grid Jan '25): Developed FMCG feature extraction pipeline using **DINOv2** + **Qwen2-VL** with PEFT (LoRA) for zero-shot tasks.
- **6th Place Globally** (ICUAS Mar'25): Built multi-drone navigation system with ArUco markers, BFS/A* and raycasting.
- **3rd Place** (IMAV Nov'23): Built autonomous lane-following drone with classical CV and PID control.

TECHNICAL SKILLS

Languages: Python, C++

ML / CV: PyTorch, TensorFlow, OpenCV, Transformers, LoRA, CLIP, YOLO, TensorRT, ONNX Runtime, OpenVINO

MLOps: Docker, CI/CD (Git, DVC), MLflow, W&B, FastAPI (REST), AWS (Lambda, EC2, S3, ECS)

Robotics: ROS, Jetson Nano, Intel NUC Ultra, RealSense D415, Raspberry Pi

Domains: Computer Vision, Deep Learning, Model Optimization, Machine Learning, Fine-Tuning